

Samarth Prabhu

Senior Data Scientist | Applied AI & Computer Vision | E-commerce

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Professional Summary

ML/AI Engineer with **5+ years** of experience and a strong foundation in **Mathematics**, turning business problems into production AI systems across **E-commerce, Marketing and Sales**. At Infosys, built ML/AI pipelines for a global marketplace cutting seller onboarding time by **40%** using AI/ML automations and achieved **90+% POC approval rate**. Equally comfortable in a model-notebook and a boardroom, turning stakeholder questions into shipped AI and shipped AI into business results recognized with a **Certificate Of Appreciation** and an **Outstanding Performance Rating**.

Work Experience

Senior ML/AI Engineer

Dec 2024 – Present

Infosys, Pune Client: Global E-commerce Marketplace

- Caught **98% of incorrect listings** (recall), by productionizing a validation service with CLIP + ML **Classifier**, **OCR** size-chart verification and **Object detection** with reason so sellers can fix and resubmit.
- Standardized ~60k+ seller images/month into brand-standard shots shipping services with an **image-shot classifier**, Image Pipelines including **BiRefNet segmentation**, shadow-preserving compositing and image transformations (re-size/ align, color-profiles) using basic **Computer Vision**
- Increased product **attribute coverage by 30%** in the pilot, by building a **Agentic workflow** on **Langraph** that extracts, infers attributes from available product data with confidence-based HITL.
- Achieved a **90+% POC approval rate** with POC's including **fine-tuned encoders** (LoRA on CLIP), **lightweight neural networks** and **LLM agents**, by presenting each solution directly to VP- and CIO-level stakeholders and taking approved work to production.

Data Scientist

Aug 2022 – Dec 2024

QuantumLink Communications, Mumbai

- Forecasted customer churn at **80% accuracy** using classification and clustering models, enabling targeted retention campaigns.
- Reduced manual sales-log triage effort by **50% using TF-IDF NLP pipelines** and **automated ETL workflows**.
- Saved 2 hours/day through automation of data ingestion and **preprocessing across 3+ data sources**.
- Built **5+ Power BI dashboards** for acquisition, retention, and engagement KPIs used in weekly leadership reviews.

Business Analyst

Jun 2021 – Feb 2022

Sciative Solutions, Navi Mumbai

- Drove **7% revenue uplift** by improving pricing strategies and market segmentation across multiple clients.
- Reduced pricing audit time by **80%** (5 hrs to 1 hr) through **Python-based profitability audit automation**.

Technical Skills

Machine Learning & Data: Regression, Classification, Clustering, Feature Engineering, NLP, Hypothesis testing

Generative AI & LLMs: RAG, Fine-tuning, Agents, MCP

Tools & Libraries: Python: Scikit-learn, Pandas, NumPy, LangChain/Langraph, Torch, SQL,

BI & Cloud: Power BI, Google Cloud Platform, FastAPI, Streamlit, git

Education

M.Sc. Mathematics

2019 – 2021

Mithibai College, Mumbai

B.Sc. Mathematics

2016 – 2019

Jai Hind College, Mumbai